scientific research, surveys, experiments etc. The reliability of the interpretation of the statistical analysis depends upon the informations collected and represented.

9.1 Time Series Analysis

Time Series analysis is one of the statistical methods used to determine the patterns in data collected for a period of time. Generally, each of us should know about the past data to observe and understand the changes that have taken place in the past and current time. One can also identify the regular or irregular occurrence of any specific feature over a time period in a time series data. Most of the time series data relates to fields like Economics, Business, Commerce, etc… For example Production of a product, Cost of a product, Sales of a product, National income, Salary of an individual, etc.. By close observation of time series data, one can predict and plan for future operations in industries and other fields.

Definition 9.1

A Time Series consists of data arranged chronologically – *Croxton & Cowden.*

When quantitative data are arranged in the order of their occurrence, the resulting series is called the Time Series – *Wessel & Wallet.*

9.1.1 Meaning, Uses and Basic Components

Meaning:

A time series consists of a set of observations arranged in chronological order (either ascending or descending). Time Series has an important objective to identify the variations and try to eliminate the variations and also helps us to estimate or predict the future values.

Why should we learn Time Series?

It helps in the analysis of the past behavior.

It helps in forecasting and for future plans.

It helps in the evaluation of current achievements.

It helps in making comparative studies between one time period and others.

Therefore time series helps us to study and analyze the time related data which involves in business fields, economics, industries, etc…

Components of Time Series

There are four types of components in a time series. They are as follows;

(i) Secular Trend

(ii) Seasonal variations

(iii) Cyclic variations

(iv) Irregular variations

(i) Secular Trend

It is a general tendency of time series to increase or decrease or stagnates during a long period of time. An upward tendency is usually observed in population of a country, production, sales, prices in industries, income of individuals etc., A downward tendency is observed in deaths, epidemics, prices of electronic gadgets, water sources, mortality rate etc…. It is not necessarily that the increase or decrease should be in the same direction throughout the given period of time.

(ii) Seasonal Variations

As the name suggests, tendency movements are due to nature which repeat themselves periodically in every seasons. These variations repeat themselves in less than one year time. It is measured in an interval of time. Seasonal variations may be influenced by natural force, social customs and traditions. These variations are the results of such factors which uniformly and regularly rise and fall

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